

Mission Files

A Slocum mission is a **plain-text file** that tells the glider what to do: where to fly, how deep to yo, which sensors to sample, when to come up for comms, and — most importantly — every condition under which it should stop what it is doing and get itself safely back to the surface. Writing and editing these files is the day-to-day craft of piloting a Slocum.

This page explains how a mission file is put together, walks through each of the standard behaviors, and shows how the same file encodes both the *work* the glider does and the *safety net* that keeps it alive. For the flip side — how to read and respond once the glider has actually aborted — see the [Aborts](#) page.

Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Section 9 — Missions and Appendix B — Code Theory and Operation / Sample Mission), the Teledyne Webb Research basic glider-training material, the `masterdata` reference, the TWR user forum (the **Missions** board in particular), and the UG2 community Slack. Behavior arguments, sensor names, and defaults change between firmware releases — **always confirm against the `masterdata` shipped with your firmware** and simulate any mission change in the lab before flying it. This is an operator's overview, not a replacement for TWR documentation or the SFMC (Slocum Fleet Mission Control) User Manual, which covers the mission-planning interface itself.

`.mi` and `.ma` files

Two file types make up a mission, and both live on the glider's flash card: